UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN
Department of Electrical and Computer Engineering

ECE 310 Digital Signal Processing – Fall 2026

Homework 2

Profs. Kamalabadi, Snyder **Due: Friday, Sep 11, 11:59pm on Gradescope**

---

1. Two systems are specified by the following input-output relations, where $x[n]$ is the input and $y[n]$ the output:

   (a) $y[n] = y[n-5] + x[n] + 10x[n-1]$

   (b) $y[n-2] + 2y[n] = \cos\left(\frac{\pi}{6}n\right)x[n]$

   (c) $y[n] = \frac{1}{2}^{|n|}x[n]$

   For each system, determine if it is: (a) linear or non-linear, (b) time-invariant or time-varying. **Justify your answers with proofs or counter-examples**.

2. The output, $y[n]$, of a given system is always related to its input $x[n]$ by $y[n] = h[n] * x[n]$, where $h[n]$ is the system's unit pulse response. Show that the system must be LSI.

3. Express the output $y[n]$ of an LSI system with unit pulse response $h[n]$ in terms of its step response $g[n] = h[n] * u[n]$ and the input $x[n]$. **Hint:** Try to represent $\delta[n]$ in terms of shifted versions of $u[n]$.

4. Assume that the zero-state response of an LSI system to input $x[n] = 3^{(-n)}u[n]$ is $y[n] = \frac{1}{5^n}u[n-1]$. Use the system's properties (linearity and shift-invariance) to find $h[n]$, the system's unit pulse response.

5. Compute the convolution $x[n] * h[n]$ for the $x[n]$ and $h[n]$ given below. **Note**: The arrow indicates $n = 0$.

   (a) $x[n] = \{-1, \underset{\uparrow}{0}, 1\}, h[n] = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$

   (b) $x[n] = 3^{(-n)}u[n], h[n] = \{\underset{\uparrow}{0}, 1, 2\}$

   (c) $x[n] = u[n], h[n] = n(u[n] - u[n-4])$

   (d) $x[n] = (-1)^{(-n)}u[n], h[n] = e^{(-n)}u[n]$

   (e) $x[n] = 0.5^n u[n], h[n] = 2^{(-n)}u[-n]$